# Shubhadeep Mandal

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## Education

**International Institute of Information Technology, Hyderabad** July 2025 – Present
*Telangana, India*
*Master of Technology in Computer Science and Engineering* *CGPA: 9.12*

**Indian Institute of Information Technology, Ranchi** Aug 2020 – Jun 2024
*Ranchi, India*
*Bachelor of Technology (Hons.) in Computer Science and Engineering* *CGPA: 9.05*

## Projects

**Negotiate, Then Solve: Multi-Step LLM Agents** | *Python, Gemini, RAG, Scikit-learn, CP-SAT*

- Built an agentic workflow that turns free-text requests from email, chat and a web portal into typed constraints, then negotiates conflicts over multiple rounds, adapting to each stakeholder reply via typed tool calls.
- Designed dual-process parsing: a calibrated TF-IDF + logistic-regression classifier fast-paths confident requests, while an LLM drafts typed constraint JSON validated deterministically.
- Built hybrid BM25 + dense-embedding retrieval with reciprocal-rank fusion over OCR-ingested policy documents, raising Recall@5 to **93.4%** (vs 87.7% BM25), with every denial required to cite a retrieved rule.
- Built an evaluation harness with baselines, oracle and ablations (McNemar, Wilcoxon, bootstrap CIs); guardrails ensured **no attack succeeded** across 50 prompt-injection, spoofing and data-leak tests.

**IndicLLM: Transformer Pretraining & Reasoning SFT** | *PyTorch, SentencePiece, ONNX Runtime Web*

- Built Hindi and Assamese LMs end to end: curated 1.25B tokens (20%+ self-crawled and OCR'd), trained native 16K BPE tokenizers, and pretrained four 25M-param Transformers from PyTorch primitives on 500M tokens each.
- Fine-tuned on solver-generated reasoning, reaching **0.96** test accuracy and **0.85** on 10-hop chains (3× the training depth) vs a 0.48 baseline, with a leakage check ruling out answer lookup.
- Exported all 10 models to ONNX with a KV cache and a JavaScript tokenizer so they run fully in the browser, deployed as a live demo on Railway via Docker.

**Football Video Analytics with YOLO** | *Python, YOLOv8, ByteTrack, OpenCV, Streamlit, K-Means*

- Fine-tuned YOLOv8 with ByteTrack for multi-object tracking of players, referees, and the ball, leveraging deterministic K-Means clustering on jersey pixels for team classification and linear interpolation for ball-gap recovery.
- Compensated camera pan, tilt and zoom with chained RANSAC pitch-plane homographies over Lucas-Kanade flow, mapping camera pixels to pitch coordinates to compute real-world sprint velocities, distance covered, and team ball possession.
- Re-architected pipeline to stream frames in batches, **cutting peak RAM from 5.3 GB to 1.0 GB** and full-video decodes from six to two; built Streamlit app with content-hashed caching for instant re-runs and stats export.

**Distributed Data Pipelines: MapReduce & gRPC Streaming** | *Hadoop, C++, Python, gRPC*

- Reduced a 1M-record Hadoop analytics job (19 outputs) to a single MapReduce stage by expressing every metric as a mergeable partial aggregate, so only compact summaries cross the shuffle; results match sequential and MPI baselines exactly.
- Architected a real-time gRPC streaming pipeline sustaining **96–123k records/s**, answering mid-ingestion queries in 2.1–4.5 ms with bounded-queue backpressure and zero record loss.

## Technical Skills

**Languages**: Python, C/C++, SQL, Bash.

**ML & Statistics**: Logistic Regression, K-Means Clustering, TF-IDF, Hybrid Retrieval, Hypothesis Testing, Bootstrap CIs.

**Deep Learning & LLMs**: PyTorch, Transformers, Hugging Face, LoRA, RAG, LLM Agents, YOLOv8, OpenCV.

**Data & Tools**: NumPy, Pandas, Scikit-learn, Hadoop MapReduce, gRPC, FastAPI, Streamlit, Linux, Git, Docker.

## Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

## Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.